

# Compiler Design (CD)

GATE Computer Science & IT Previous Year Questions (2016 – 2025) | Total Questions: 13

## Question 1 of 13 | GATE 2025 Set 1 (CS1) | Q.12 (Computer Science & IT)

Q.12

Which ONE of the following statements is FALSE regarding the symbol table?

- (A) Symbol table is responsible for keeping track of the scope of variables.
- (B) Symbol table can be implemented using a binary search tree.
- (C) Symbol table is not required after the parsing phase.
- (D) Symbol table is created during the lexical analysis phase.

## Question 2 of 13 | GATE 2025 Set 1 (CS1) | Q.13 (Computer Science & IT)

Q.13

Which ONE of the following techniques used in compiler code optimization uses live variable analysis?

- (A) Run-time function call management
- (B) Register assignment to variables
- (C) Strength reduction
- (D) Constant folding

## Question 3 of 13 | GATE 2025 Set 1 (CS1) | Q.46 (Computer Science & IT)

Q.46

Which of the following statement(s) is/are TRUE while computing First and Follow during top down parsing by a compiler?

- (A) For a production  $A \rightarrow \epsilon$ ,  $\epsilon$  will be added to  $\text{First}(A)$ .
- (B) If there is any input right end marker, it will be added to  $\text{First}(S)$ , where  $S$  is the start symbol.
- (C) For a production  $A \rightarrow \epsilon$ ,  $\epsilon$  will be added to  $\text{Follow}(A)$ .
- (D) If there is any input right end marker, it will be added to  $\text{Follow}(S)$ , where  $S$  is the start symbol.

## Question 4 of 13 | GATE 2025 Set 1 (CS1) | Q.52 (Computer Science & IT)

Q.52

Refer to the given 3-address code sequence. This code sequence is split into basic blocks. The number of basic blocks is \_\_\_\_\_. (Answer in integer)

1001:  $i = 1$

```
1002: j = 1
1003: t1 = 10*i
1004: t2 = t1+j
1005: t3 = 8*t2
1006: t4 = t3-88
1007: a[t4] = 0.0
1008: j = j+1
1009: if j <= 10 goto 1003
1010: i = i+1
1011: if i <= 10 goto 1002
1012: i = 1
1013: t5 = i-1
1014: t6 = 88*t5
1015: a[t6] = 1.0
1016: i = i+1
1017: if i <= 10 goto 1013
```

[ Numerical Answer Type (NAT) Question — Enter Numerical Answer ]

**Question 5 of 13 | GATE 2025 Set 2 (CS2) | Q.21 (Computer Science & IT)**

Q.21

Consider the following statements about the use of backpatching in a compiler for intermediate code generation:

(I)

Backpatching can be used to generate code for Boolean expression in one pass.

(II)

Backpatching can be used to generate code for flow-of-control statements in one pass.

Which ONE of the following options is CORRECT?

- (A) Only (I) is correct.
- (B) Only (II) is correct.
- (C) Both (I) and (II) are correct.
- (D) Neither (I) nor (II) is correct.

**Question 6 of 13 | GATE 2025 Set 2 (CS2) | Q.22 (Computer Science & IT)**

Q.22

Given the following syntax directed translation rules:

Rule 1:  $R \rightarrow AB$  {B. i = R. i - 1; A. i = B. i; R. i = A. i + 1; }

Rule 2:  $P \rightarrow CD$  {P. i = C. i + D. i; D. i = C. i + 2; }

Rule 3:  $Q \rightarrow EF$  {Q. i = E. i + F. i; }

Which ONE is the CORRECT option among the following?

- (A) Rule 1 is S-attributed and L-attributed; Rule 2 is S-attributed and not L-attributed; Rule 3 is neither S-attributed nor L-attributed
- (B) Rule 1 is neither S-attributed nor L-attributed; Rule 2 is S-attributed and L-attributed; Rule 3 is S-attributed and L-attributed
- (C) Rule 1 is neither S-attributed nor L-attributed; Rule 2 is not S-attributed and is L-attributed; Rule 3 is S-attributed and L-attributed
- (D) Rule 1 is S-attributed and not L-attributed; Rule 2 is not S-attributed and is L-attributed; Rule 3 is S-attributed and L-attributed

**Question 7 of 13 | GATE 2024 Set 1 (CS1 S5) | Q.26 (Computer Science & IT)**

Q.26

Which of the following is/are Bottom-Up Parser(s)?

- (A) Shift-reduce Parser
- (B) Predictive Parser
- (C) LL(1) Parser
- (D) LR Parser

**Question 8 of 13 | GATE 2024 Set 1 (CS1 S5) | Q.37 (Computer Science & IT)**

Q.37

Consider the following syntax-directed definition (SDD).

$S \rightarrow DHTU$

$\{ S.val = D.val + H.val + T.val + U.val; \}$

$D \rightarrow "M"D1$

$\{ D.val = 5 + D1.val; \}$

$D \rightarrow \epsilon$

$\{ D.val = -5; \}$

$H \rightarrow "L"H1$

$\{ H.val = 5 \times 10 + H1.val; \}$

$H \rightarrow \epsilon$

$\{ H.val = -10; \}$

$T \rightarrow "C"T1$

$\{ T.val = 5 \times 100 + T1.val; \}$

$T \rightarrow \epsilon$

$\{ T.val = -5; \}$

$U \rightarrow "K"$

$\{ U.val = 5; \}$

Given "MMLK" as the input, which one of the following options is the CORRECT value computed by the SDD (in the attribute S.val)?

- (A) 45
- (B) 50
- (C) 55
- (D) 65 Organizing Institute: IISc Bengaluru

**Question 9 of 13 | GATE 2024 Set 1 (CS1 S5) | Q.39 (Computer Science & IT)**

Q.39

Consider the following pseudo-code.

L1:  $t1 = -1$ L2:  $t2 = 0$ L3:  $t3 = 0$ L4:  $t4 = 4 \square t3$ L5:  $t5 = 4 \square t2$ L6:  $t6 = t5 \square M$ L7:  $t7 = t4 + t6$ L8:  $t8 = a[t7]$ L9: if  $t8 \leq \text{max}$  goto L11L10:  $t1 = t8$ L11:  $t3 = t3 + 1$ L12: if  $t3 < M$  goto L4L13:  $t2 = t2 + 1$ L14: if  $t2 < N$  goto L3L15:  $\text{max} = t1$ 

Which one of the following options CORRECTLY specifies the number of basic blocks and the number of instructions in the largest basic block, respectively ?

- (A) 6 and 6
- (B) 6 and 7
- (C) 7 and 7
- (D) 7 and 6 Organizing Institute: IISc Bengaluru

**Question 10 of 13 | GATE 2024 Set 2 (CS2 S6) | Q.21 (Computer Science & IT)**

Q.21

Consider the following two sets:

Set X Set Y

P. Lexical Analyzer 1. Abstract Syntax Tree

Q. Syntax Analyzer 2. Token

R. Intermediate Code Generator 3. Parse Tree

S. Code Optimizer 4. Constant Folding

Which one of the following options is the CORRECT match from Set X to Set Y ?

- (A) P – 4; Q – 1; R – 3; S – 2
  - (B) P – 2; Q – 3; R – 1; S – 4
  - (C) P – 2; Q – 1; R – 3; S – 4
  - (D) P – 4; Q – 3; R – 2; S – 1
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**Question 11 of 13 | GATE 2023 Single Session | Q.36 (Computer Science & IT)**

Q.36

CS

Consider the control flow graph shown.

$j = n$

$a = 10$

$i = i + 1$

$j = j - 1$

B1

B2

$i = a + 1$

$a = 20$

B3

B4

EXIT

ENTRY

$i = m - 1$

Which one of the following choices correctly lists the set of live variables at the exit point of each basic block?

- (A) B1: {}, B2: {a}, B3: {a}, B4: {a}
  - (B) B1: {i, j}, B2: {a}, B3: {a}, B4: {i}
  - (C) B1: {a, i, j}, B2: {a, i, j}, B3: {a, i}, B4: {a}
  - (D) B1: {a, i, j}, B2: {a, j}, B3: {a, j}, B4: {a, i, j}
- CS

**Question 12 of 13 | GATE 2019 Single Session | Q.3 (Computer Science & IT)**

Q.3

Which one of the following kinds of derivation is used by LR parsers?

- (A) Leftmost
- (B) Leftmost in reverse
- (C) Rightmost

(D) Rightmost in reverse

**Question 13 of 13 | GATE 2016 Set 1 (CS1) | Q.46 (Computer Science & IT)**

Q.46

Consider the following Syntax Directed Translation Scheme (SDTS), with non-terminals

{S, A} and terminals {a, b}.

S

→

aA

{ print 1 }

S

→

a

{ print 2 }

A

→

Sb

{ print 3 }

Using the above SDTS, the output printed by a bottom-up parser, for the input aab is:

(A) 1 3 2

(B) 2 2 3

(C) 2 3 1

(D) syntax error